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Molecular dynamic study on the transport properties of ionic liquids in ZTC porous carbon materials

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ABSTRACT

The adsorption and diffusion behavior of ions within electrodes have a significant impact on the energy density and charging/discharging speed of supercapacitors. Zeolite-Templated Carbon (ZTC) material, a type of porous carbon material characterized by well-defined pore shape and spatial structure, is recognized as a promising electrode material owing to its numerous advantages. This study employs molecular dynamics simulations to investigate the diffusion and adsorption profiles of ionic liquids [EMI][BF4] in different types of ZTC materials as electrodes. Additionally, the effect of ion size, pore dimension, pore shape, and the properties of the adsorbed ions are discussed. The degree of confinement (DOC) and ion diffusivity are thoroughly analyzed to elucidate the transport behavior of ions within the ZTC pores. The results demonstrate that the diffusion coefficient and degree of confinement of ionic liquids predominantly depend on the anion size and the pore size of the porous carbon. Surprisingly, in certain cases, larger anions exhibit enhanced diffusion ability and adsorption effect. These findings unravel the underlying transport mechanism of ions in ZTC materials and provide valuable theoretical insights for the application of ZTC materials as electrodes in the field of supercapacitors.

1. Introduction

The supercapacitor is a promising, environmentally friendly energy storage device that bridges the gap between batteries and conventional capacitors. It comprises three key components: positive and negative electrodes, electrolyte, and separator [1,2]. Feng et al. [3] made a comprehensive review on the electrode materials, and electrolytes for high-frequency micro-supercapacitors used as alternating current line filters to rectify pulse energy. Supercapacitors can be categorized into two types based on their energy storage mechanisms: double-layer supercapacitors and pseudocapacitors [4–6]. In double-layer supercapacitors, electrical energy is stored and released through the formation and detachment of a double layer of co-ions and counter-ions, which is only a few nanometers thick and forms on the electrode walls. Electrode materials play an essential role in determining the whole electrochemical charge storage performance. Porous carbon electrodes are considered to be highly promising for commercial double-layer supercapacitors due to their advantages of high specific surface area, excellent electrical conductivity, and low cost [7]. Several types of carbon electrode materials have been proposed and studied, including activated carbon [8], carbon nanotubes [9,10], carbon sheets [11], carbide derivatives [12], onion carbon [13], graphene [14,15] and so on. Yang et al. [16] reviewed the application progress of nanocarbon-based materials and nanocarbon-based composite materials as electrode materials in supercapacitors.

For regularly shaped carbon materials such as carbon nanotubes, Liu et al. [17–19] provided a thorough analysis of the water diffusing process. However, the disordered pore structure of most commonly used carbon materials poses challenges in accurately understanding the adsorption behavior of ions within the pores. Recently, the ZTC material, a type of porous carbon material, has gained significant attention. ZTC material is synthesized through the chemical vapor deposition process of carbon-containing precursors on zeolite templates, resulting in a regular microporous carbon structure composed of graphene. Unlike other porous carbon materials, the organized nanopore architecture in ZTC materials facilitates easier ion diffusion. ZTC materials exhibit a distinctive one- to three-dimensional microporous structure, where the pores are interconnected and traversed by pore channels. This microporous structure directly influences the physical and chemical properties of ZTC materials, determining the size, orientation, and shape of the

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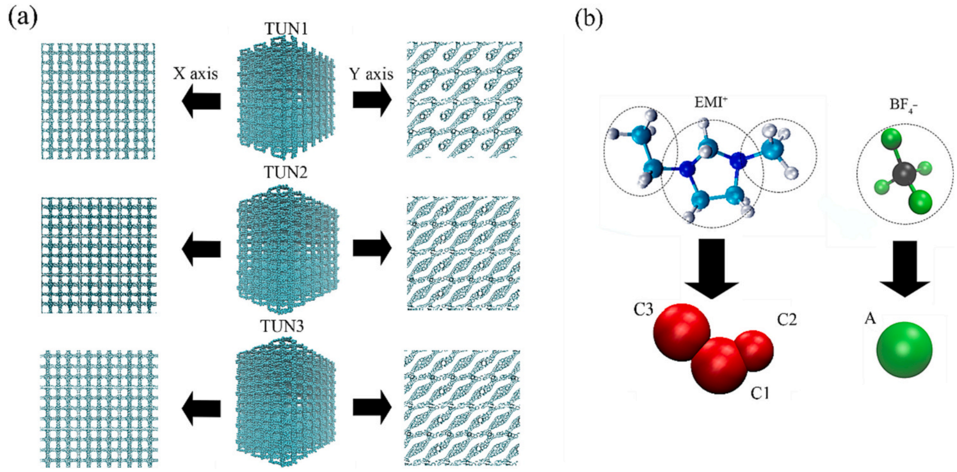

Fig. 1. (a) Molecular structure of the ZTC materials, and (b) Coarse-grained model of ionic liquids [EMI][BF4].

pore channels, as well as the orientation and shape of the microporous structures. Due to their unique porous structure, different types of ZTC materials can be classified according to the pore size, pore shape, and pore connectivity. For example, the pores in ZTC are divided roughly into small-pore (with pore size smaller than 4 Å), medium-pore (with pore size between 5–10 Å) and large-pore (with pore size greater than 12 Å). According to the dimension of the uniformity pore channel, the microchannel in the ZTC materials can be divided into one-dimensional channel, two-dimensional channel and three-dimensional channel. [20–22] One significant advantage of ZTC materials is that the pore structure can be tailored by substituting different zeolite templates, enabling simulation studies with various material classes and straightforward chemical modifications. Lahrar et al. [23] found that the carbon with the ordered structure, and a well defined pore size, has a much higher capacitance than the carbon with disordered structure and a broader pore size distribution.

The diffusion and adsorption properties of ions in the electrode material significantly impact the energy storage capacity and overall efficiency of supercapacitors. Understanding the relationship between ion transport properties, porous carbon structure, and the electrochemical performance of supercapacitors remains a remarkable challenge. Borghi et al. [24] conducted experimental studies that revealed solid-like structures within the porous carbon framework and a double-layer structure at the solid-liquid interface. These findings indicated that the presence of the solid-like phase at the interface affects the charging process of ionic liquids in nanoporous carbon-based planar supercapacitors. Zhang et al. [25] investigated the variation of pore size in Al-MOF single crystal-derived porous carbon to determine the most suitable pore size and employed nuclear magnetic resonance techniques to elucidate the charge storage mechanism of porous carbon at the microscopic level. Lahrar et al. [26] utilized simulations to explore the micro-mechanisms of ion adsorption in porous carbon structures. Liu et al. [27] applied molecular dynamics simulations to reveal the effects of membrane porosity on interfacial resistance in carbon nanotube membranes. Additionally, the influence of pore size on hydrated $Zn^{2}+$ ion storage in nanostructured carbon with divergent pore architectures is investigated by experiment and molecular dynamic simulation [28]. These studies have provided valuable insights into the charge storage mechanism of ionic liquids in double-layer supercapacitors. However, research on ion adsorption properties in porous carbon structures is limited, with most studies focusing on a single porous carbon structure. The correlation between ion size, porous structure, and interfacial properties is still unclear. Experiments are limited by the complexity and preservation challenges associated with preparing ionic liquids. Molecular dynamic simulations offer a promising avenue to overcome these limitations and provide a more accurate understanding of the dynamic transport and adsorption processes of ions in porous carbon structures.

To our knowledge, the impact of the pore structure in the porous carbon nanomaterial ZTC [29] on the diffusion and adsorption characteristics of ionic liquids has not been fully elucidated. In this paper, molecular dynamics simulations are adopted to explore various adsorption models of ionic liquids in the ZTC porous carbon electrodes. The adsorption and diffusion behaviors of ions through the degree of confinement (DOC) and ion diffusion rate are also characterized. Additionally, the effects of pore shape, pore size, and ion species on the ion transport and adsorption properties are discussed.

2. The details of molecular simulation

This paper investigates the adsorption properties of ionic liquids using three distinct types of porous ZTC materials, namely TUN1, TUN2, and TUN3, as depicted in Fig. 1(a). These designations were chosen to facilitate clear and precise descriptions. The three TUN models exhibit similar pore structures, allowing the exploration of the effect of pore geometry on the ion diffusion. To characterize the porous structure, Lev Sarkisov's [30] method is used to calculate the pore size distribution (PSD) in the ZTC materials. The commonly used ionic liquids [EMI][BF4] are applied as the electrolyte [31]. Considering computational efficiency while ensuring simulation accuracy, a coarse-grained model was adopted to represent the [EMI][BF4] ion. As illustrated in Fig. 1(b), the cation is represented by three distinct coarse particles (C1, C2, C3), while the anions are rendered by a single coarse particle (A). For detailed information on the coarse-grained model and associated parameters, please refer to the previous work by Merlet [32]. Table 1 also

Table 1
Force field parameters of the coarse particles for [EMI][BF4].

Particles	x (Å)	y (Å)	z (Å)	M (g/mol^{-1})	ε_i (kcal/mol^{-1})	σ_i (Å)	q_i (e)
C1	0.000	−0.527	1.365	67.07	2.56	4.38	0.3591
C2	0.000	1.641	2.987	15.04	0.36	3.41	0.1888
C3	0.000	−0.737	−1.653	29.07	1.24	4.38	0.2321
A	0.000	0.000	0.000	86.81	3.24	4.51	−0.78

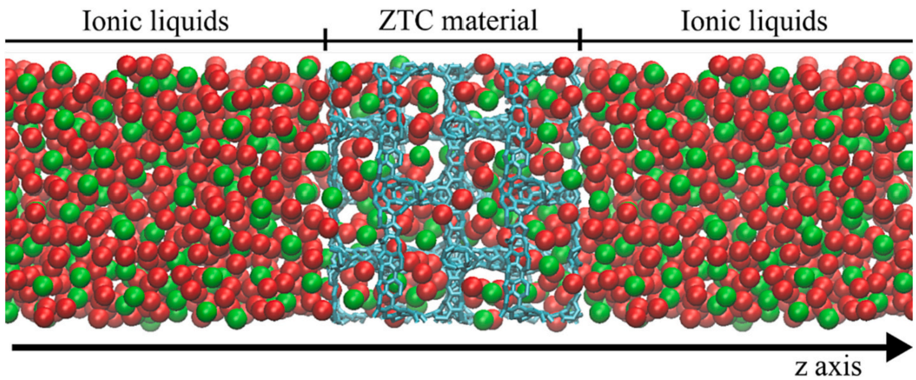

Fig. 2. Molecular dynamics simulation system of pure ionic liquids diffusing in porous carbon. Anions are represented in green, cations in red, and carbon atoms in light blue. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

summarizes the specific parameters for these coarse particles.

As reported by Crespo and Vazquez [33,34], the simulation results of the coarse-grained model have demonstrated excellent agreement with experimental findings. Furthermore, in our previous study, this coarse-grained model was effectively utilized to investigate wetting properties [35–37]. To calculate the intermolecular van der Waals force and electric field forces between coarse particles and carbon atoms, the classic 12–6 Lennard-Jones potential combined with the Coulomb law is employed [38],

$$u_{ij}(r_{ij}) = 4\varepsilon_{ij}\left[\left(\frac{\sigma_{ij}}{r_{ij}}\right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}}\right)^{6}\right] + \frac{q_i q_j}{4\pi\varepsilon_0} \tag{1}$$

where q_i denotes the charge of the i atom, r_{ij} is the distance between the i and j atoms, ε_{ij} represents the energy parameter, and σ_{ij} is the scale parameter for atoms i and j, the short-range LJ force is zero when the distance between atoms i and j is σ_{ij}.

In this paper, all simulations were carried out using the LAMMPS software [39]. Fig. 2 illustrates the simulation model depicting the diffusion of ionic liquids in porous ZTC materials. The ZTC materials were positioned at the middle of the simulation box, while the left and right sides were filled with ionic liquids. The size of the simulation box was set to 30 Å × 40 Å × 91 Å. To reduce computational time, the ZTC material was immobilized, allowing the particles of the ionic liquid to freely diffuse into the pores from the exterior of the material. A cutoff radius of 15 Å was applied for the Lennard-Jones interaction. The long-range electrostatic Coulomb interaction between coarse particles was calculated using the particle-particle-particle-mesh (PPPM) method [40,41], with an accuracy of 10^{-4}. The whole system consisted of 1200 ion pairs of ionic liquids. Initially, simulations were conducted in the NVT ensemble at a temperature of 298 K for 0.5 ns with a time step of 0.1 fs. Subsequently, the time step was increased to 1 fs, and the system was run for 3 ns at 298 K to allow complete and unrestricted diffusion of ions, enabling the analysis of ion adsorption and diffusion profiles in different ZTC materials. The simulation was repeated three times to ensure the accuracy of the results.

To elucidate the diffusion and adsorption characteristics of ionic liquids within the porous carbon structure, several key parameters are calculated during the simulation, including the diffusion coefficient, diffusion rate, and DOC. In molecular dynamics, the mean square displacement (MSD) serves as a crucial indicator for describing atomic motion. A higher MSD value indicates more intense molecular motion and potentially lower thermal stability of the system. The calculation of MSD follows the formula [42]:

$$MSD = \left\langle |r_i(t) - r_i(0)|^2 \right\rangle \tag{2}$$

where $r_i(t)$ and $r_i(0)$ represent the position of atom i at moment $t = t$ and $t = t_0$, and the pointed bracket denotes the mean value. According to the definition of MDS, the slope of the mean square displacement with respect to time provides information about the diffusion coefficient of ionic liquids. By employing the well-known Einstein relation [43,44] and examining the long time limit of MSD, the diffusion coefficient can be determined. The calculation procedure for obtaining the diffusion coefficient is outlined as follows,

$$D = \frac{1}{6}\lim_{t\to\infty}\frac{\left\langle |\vec{r}(t_0+t) - \vec{r}(t_0)|^2 \right\rangle}{t} \tag{3}$$

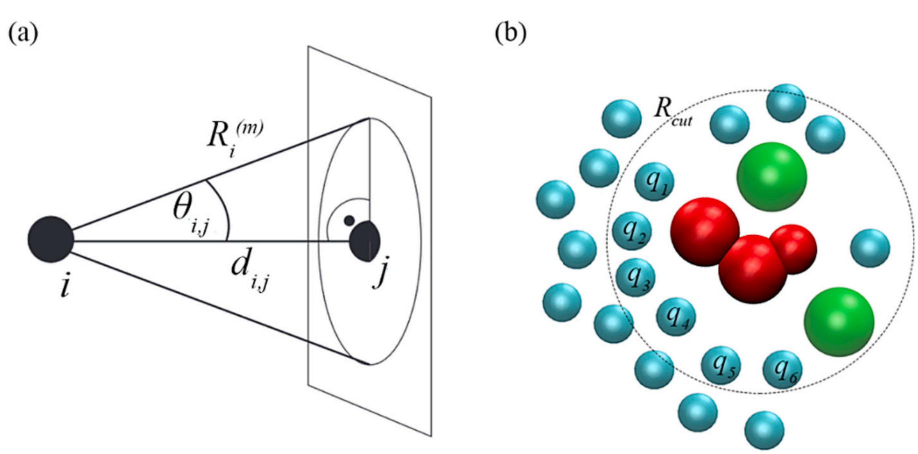

Fig. 3. (a) The definition of solid angle, and (b) Local structure characterization. Around each molecular ion, the number of neighbors of each kind of species is determined using a spherical cutoff distance R_{cut}.

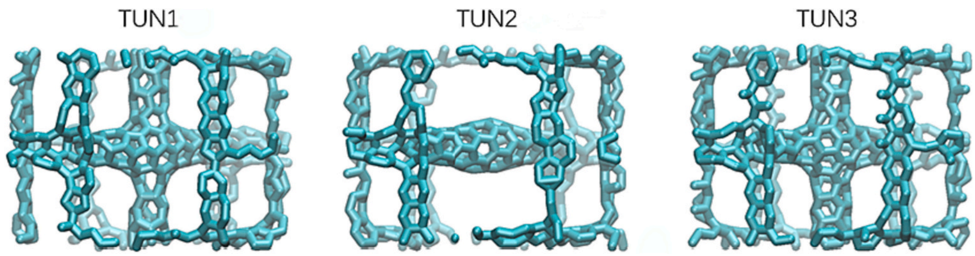

Fig. 4. Detailed diagram of the single topology of 3 TUN type ZTC materials.

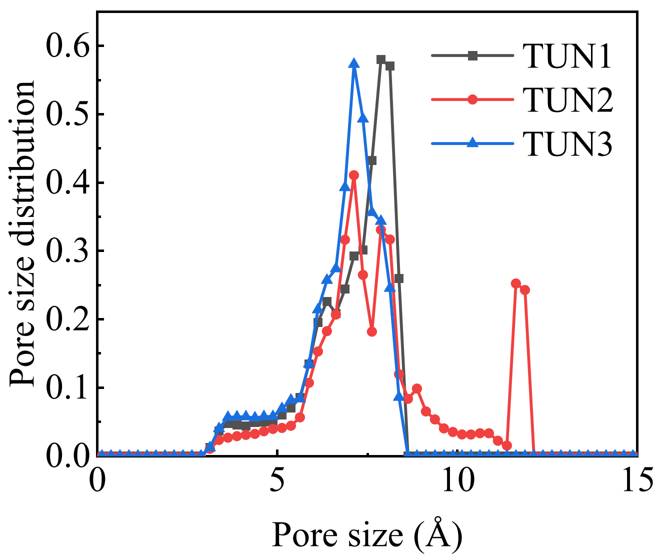

Fig. 5. The geometric pore size distribution of the three materials.

To accurately capture the variations in ion diffusion in the ZTC materials, the diffusion rate is defined as the rate at which the number of ions per unit volume increases over time. The calculation formula for the diffusion rate is as follows,

$$\nu = \frac{1}{V}\frac{dN}{dt} \tag{4}$$

Besides, to characterize the adsorption state of ions within the ZTC materials, the DOC was calculated. Previous studies by Jacobus A and Merlet [45,46] have defined the DOC as the percentage of the solid angle surrounding the ions that are occupied by the carbon atoms, normalized by the maximum value this quantity can take. For a given ion i, the solid angle associated with a carbon atom j, located at a distance d_{ij} from the ion i, is determined using the following equation,

$$\alpha_{ij} = 2\pi\left[1 - cos\left(\theta_{ij}\right)\right] = 2\pi\left(1 - \frac{d_{ij}}{\sqrt{d_{ij}^2 + R_j^2}}\right) \tag{5}$$

As shown in Fig. 3, R_j represents the radius of the region occupied by j. α_{ij} is the solid angle between a single particle and an ion. When the ion is completely enveloped, the sum of all solid angles associated with adjacent particles equals 4π. The DOC for a specified ion i is calculated as follows,

$$\frac{\sum_{j=1}^{CN}\alpha_{ij}}{4\pi} \times 100 \tag{6}$$

where CN is the number of carbon atoms fitting around ion i. DOC depends on both the number of carbon atoms around the ion and the distance of each ion from the carbon atom.

3. Results and discussion

3.1. Effect of pore geometry on ion diffusion and adsorption

To examine the influence of pore geometry on the adsorption properties of ions, the ZTC materials generated by Braun [29] are used as the porous substrates. Among these materials, the TUN-configuration ZTC materials with similar pore sizes and pore structures are chosen to minimize the impact of these factors. Fig. 4 provides a detailed illustration of the individual topologies of the three TUN-type ZTC materials. Clearly, these materials exhibit comparable pore structures. Notable, the TUN2-type material lacks a carbon chain in the middle, resulting in a smoother diffusion channel for ions along the z-axis. To provide a clear description of these materials, the geometric pore size distributions

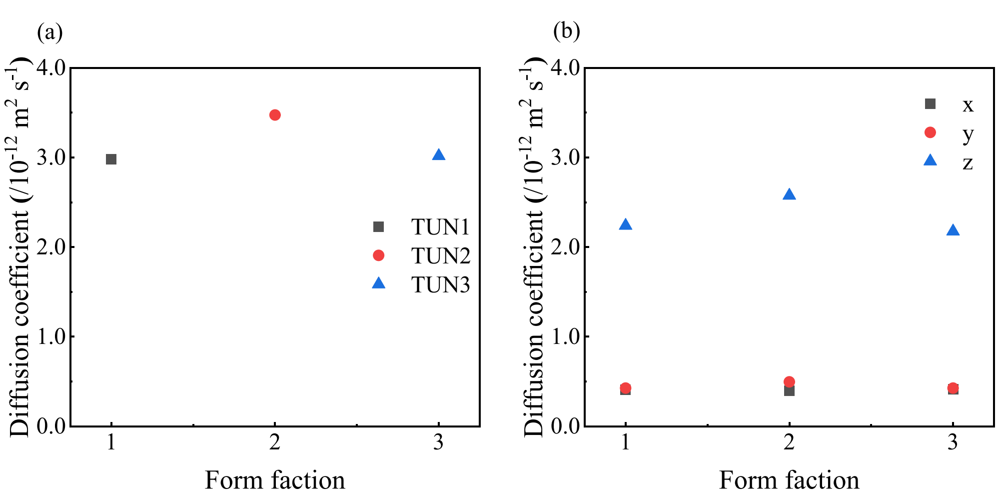

Fig. 6. Diffusion coefficient of ions in the three types of ZTC materials (a) the average value (b) the specific value in the x, y, and z-axis directions.

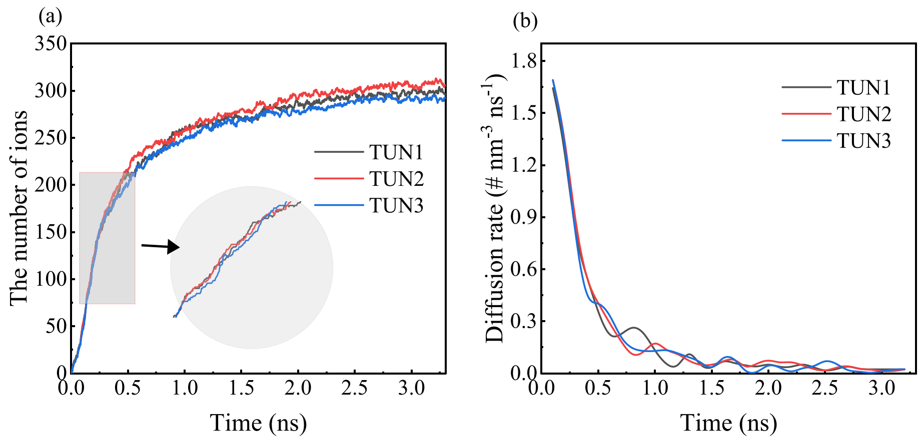

Fig. 7. Time evolution of the number of ions adsorbed in the three types of ZTC materials and the corresponding diffusion rate of ions in the three materials.

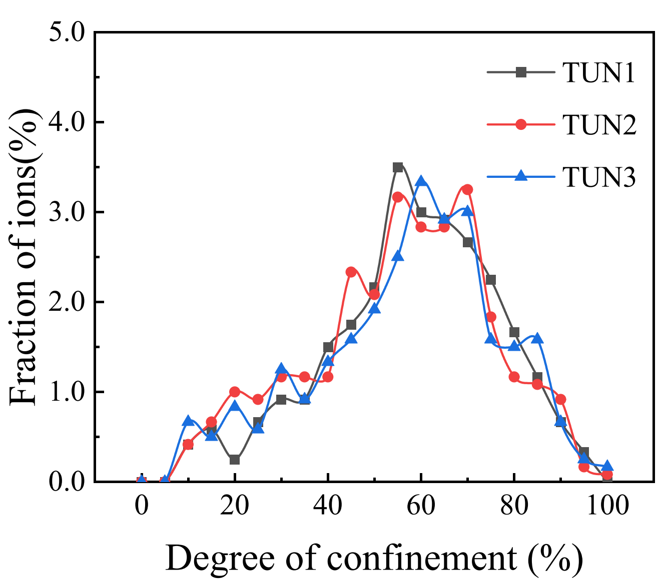

Fig. 8. DOC distribution of ions in the three of ZTC materials.

(PSD) of the three ZTC materials are calculated and shown in Fig. 5. The definition and calculation details of PSD have been given in literature [30]. It is evident that the pore sizes of TUN1 and TUN3 materials are mainly between 5 Å and 8 Å. In contrast, the pore sizes of TUN2 materials exhibit a broader distribution, encompassing pore sizes between 5 Å and 10 Å. It is worth noting that the TUN2 materials also shows a pore size peak at 11–12 Å because it lacks a carbon chain in the middle, as shown in Fig. 4. Additionally, the specific surface area of these three materials are calculated using the tools as discussed in literature [30] and it is 2369.14, 2415.91, and 2142.58 m^2/g for the TUN1, TUN2 and TUN3, respectively.

Fig. 6(a) illustrates the diffusion coefficient of the same ionic liquids in the three TUN configurations of the materials. It is evident that the TUN2 material exhibits the highest diffusion coefficient among them, while the diffusion coefficients in TUN1 and TUN3 are approximately equal. Examining the ion diffusion in distinct directions, Fig. 6(b) reveals that the diffusion coefficient of the ionic liquids in the x, y, and z-axis directions follows the same pattern as the overall diffusivity. Notably, the TUN2 material consistently possesses the largest diffusion coefficient. In terms of individual directions, the z-axis direction demonstrates the highest diffusion coefficient, while the diffusion coefficients in the x and y axes are 7 to 8 times smaller than that in the z-axis; however, the disparity between the diffusivities in the x and y axes is not significant. This observation indicates that the difference in the diffusivity of ionic liquids along the z-axis is the main factor for the higher diffusion coefficient in the TUN2 material compared to the other materials. To depict the diffusion rate of ionic liquid in the ZTC material, Fig. 7 presents the number of ions adsorbed in the three types of ZTC materials and the corresponding diffusion rate of ions. It is clear that the diffusion rate of the ionic liquid is higher before 1 ns and gradually diminishes thereafter, eventually approaching zero. The similarity in the amplitudes of the three curves proves that the difference in diffusion velocity of ionic liquids in the three materials is minimal.

When examining the adsorption of ions in porous carbon materials, a crucial aspect to consider is determining the actual quantity of ions adsorbed within the pores. This information provides insights into the adsorption capacity of ions in the ZTC material. Fig. 7(a) depicts the number of ions present in the ZTC material throughout the simulation. Similar to the ion diffusion rate graph, the number of ions in the ZTC material experiences rapid growth within the initial 0.5 ns, followed by a gradual approach to equilibrium after 2 ns. The number of ions stabilizes within the range of 270 and 300, with the TUN2 material exhibiting the highest number of ions and the TUN3 material displaying the lowest. To better understand the variations in ion adsorption characteristics among different materials, Fig. 8 illustrates the distribution of DOC for ions in TUN materials. Although the DOC distribution does not provide specific details about ion adsorption in terms of atomic environments encountered during the process, it showcases the adsorption state of ions in ZTC materials, enabling a deeper understanding of their adsorption characteristics. Notably, the red line representing the TUN2 material exhibits a more pronounced and symmetrically bimodal distribution with obvious peak at 55 % and 70 %. The blue line representing the TUN3 material shows a asymmetrically bimodal distribution with a obvious peak at 57 % and a small peak at 70 %. While, the curve for the TUN1 material exhibits a single peak distribution at 60 %. Overall, the peaks of the DOC distribution for all three materials fall within the 50 % to 70 % range, indicating that the ions mostly exist in a semi-enveloped state.

In addition, the curve representing TUN2 reveals a small peak at 45 %, suggesting that the TUN2 material offers more planar adsorption sites, attracting a greater number of [EMI][BF4] ions compared to the other materials. This observation aligns with the number of adsorbed ions depicted in Fig. 7(a). The intricate pore geometry of the TUN2 material, along with the increased interconnectivity among pores, provides more transport channels, enhancing ion diffusion capabilities and resulting in a higher number of adsorbed ions. However, it is important to note that the relationship between pore size, inter-pore channels, and the number of adsorbed ions is not a straightforward one, and this will be further discussed in subsequent sections.

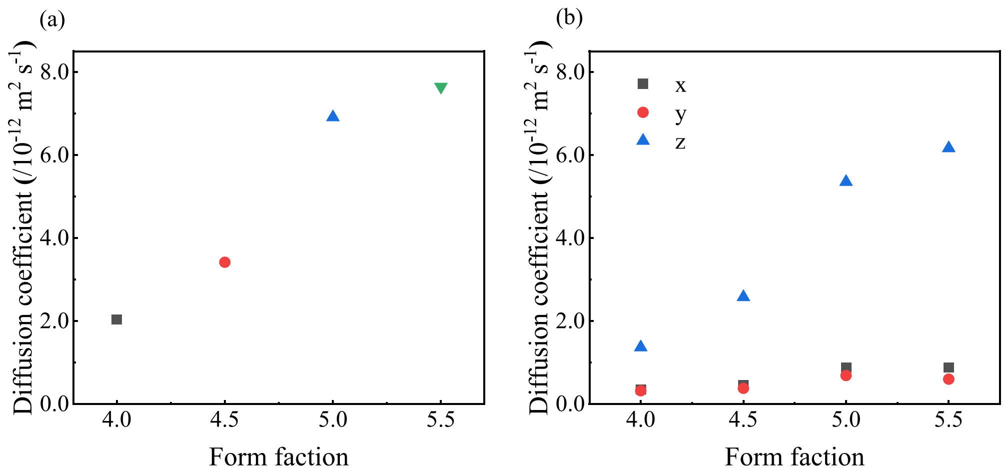

Fig. 9. Diffusion coefficient of ions in the three types of ZTC materials (a) the average value (b) the specific value in the x, y, and z-axis directions.

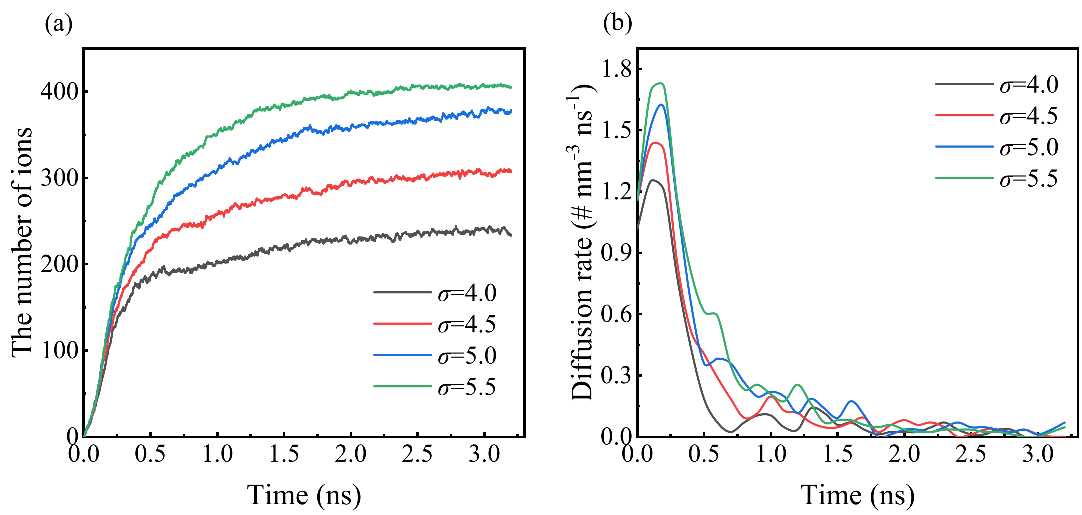

Fig. 10. Time evolution of the number of ions adsorbed in the three types of ZTC materials and the corresponding diffusion rate of ions in the three materials.

3.2. *Effect of anion size on the diffusion and adsorption*

To explore how the size of ions affects the diffusion and adsorption properties within ZTC materials, this section focuses on the TUN2 material, known for its favorable ion diffusion characteristics in the preceding subsection. For the purpose of in-depth understanding and

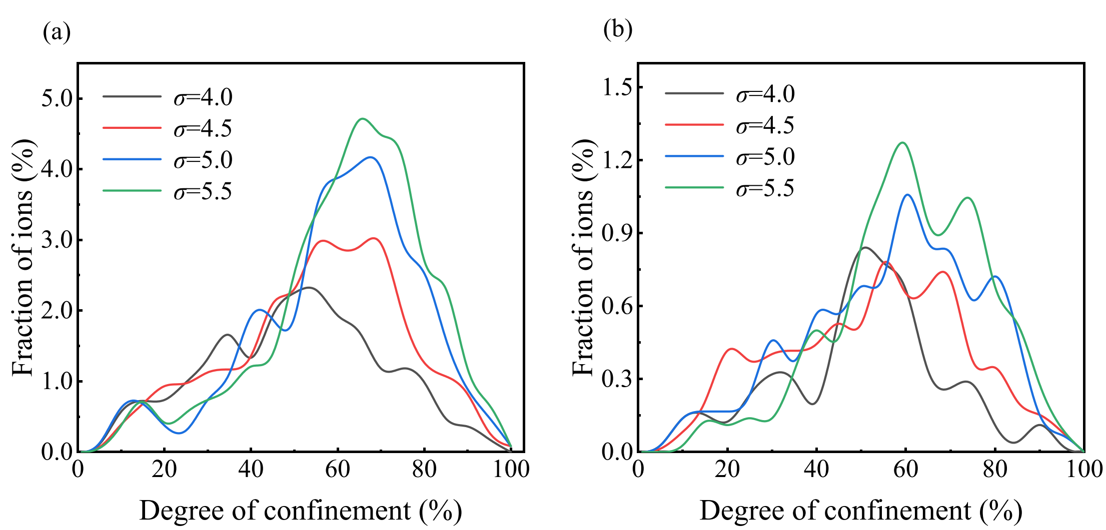

Fig. 11. DOC distribution of (a) all ions and (b) anions in the cases with different anion sizes.

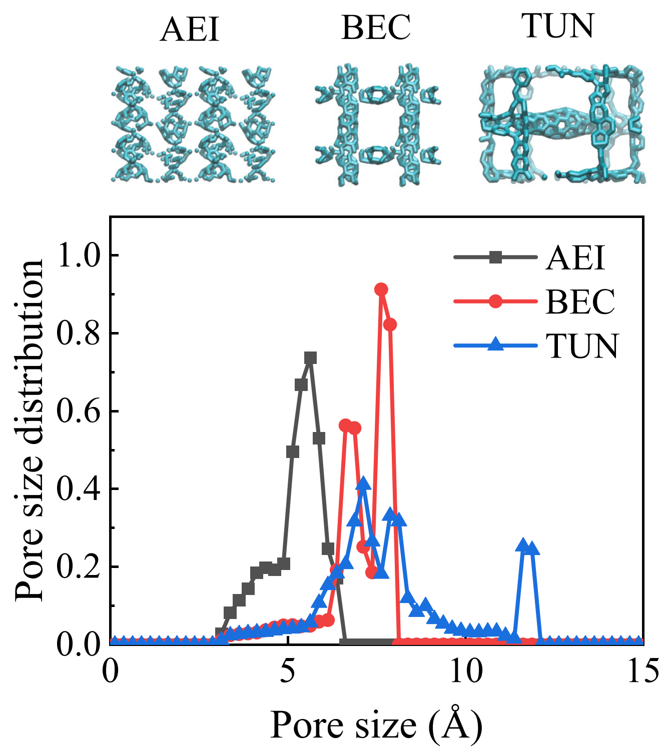

Fig. 12. Structural diagram of the AEI, BEC, and TUN materials, (b) the diffusion coefficient of ions in the three materials, and (c) geometric pore size distribution of the three materials.

computational convenience, only the σ parameter of the anion in the adopted Lennard-Jones potential during the simulation. To facilitate this analysis, a coarse-grained model of ionic liquids is employed, where the anion is represented as a single spherical site. Building upon the original model proposed by Merlet [32], the simulations for four different cases, each featuring an anion size of 4.0 Å, 4.5 Å, 5.0 Å, and 5.5 Å are conducted, respectively.

In Fig. 9, the diffusion coefficient of ionic liquids within the TUN2 material is displayed for varying anion sizes. Notably, the maximum diffusion coefficient of ions reaches 7.6×10^{-12} m^2 s^{-1} for an anion size of 5.5 Å. Overall, the diffusivity of ions demonstrates an increasing trend as the anion size increases. Particularly noteworthy is the substantial increase in ion diffusivity observed when the anion size transitions from 4.5 Å to 5.0 Å. Furthermore, Fig. 10 illustrates the number of ions adsorbed in the three types of ZTC materials and the corresponding diffusion rate of ions, which also shows a similar pattern of increasing with larger anion sizes. When considering these observations collectively, the rise in the diffusion coefficient of ionic liquids may be related to a decrease in density as the anion size increases. The rise in ion-ion distance reduces electrostatic interactions, thereby contributing to enhanced diffusivity. In addition, the range of ion sizes between 4.5 Å and 5.0 Å may have surpassed a critical threshold, leading to greater contact between ions and the ZTC material. The frequent interaction between ions and the porous material, along with mutual collisions, results in a significant boost in diffusivity.

In the context of ion adsorption performance, the influence of ion size can be seen in Fig. 10(a). It becomes evident that the number of ions reaching equilibrium within the ZTC material increases as the ion size grows. The change in anion size has a dramatic effect on ion adsorption, with the number of ion pairs shifting from approximately 200 pairs for $\sigma = 4.0$ Å to over 400 pairs for $\sigma = 5.5$ Å, indicating a substantial increase in ion adsorption. The DOC distribution of ionic liquids with different anion sizes in the TUN2 material is shown in Fig. 11(a). For smaller anions with $\sigma = 4.0$ Å, the peak value of DOC is observed at about 50 %. As the anion size increases, the peak gradually shifts towards higher values, while the maximum value of each peak also increases, implying a greater number of ions being adsorbed by the material. This trend signifies a transition from flat plate adsorption to semi-enveloped adsorption. To focus solely on the anions and exclude cation interference, the DOC distribution of the anions is calculated separately, as demonstrated in Fig. 11(b). The overall trend remains evident, with anions size of 4.0 Å displaying a peak at 50 %. Subsequently, as the anion size increases, the peak of the DOC curve progressively becomes larger. The result is consistent well with the ions adsorption phenomenon in one dimensional pores reported by Ali Eftekhari [47], that is, the ion adsorption capacity of a material is stronger when the pore size matches the ion size. According to the result in this work, ion adsorption also follows the same regularity in TUN materials with more complex pore structures. This phenomenon can be attributed to the larger pores and complex overall pore structure of the TUN material, facilitating the adsorption of larger ions in close contact with the pore walls. Conversely, smaller ions encounter difficulties in adsorption due to their inability to form a suitable fit within the pores. This finding provides important insights for understanding the behavior of ions in complex porous materials.

3.3. Effect of pore size on ion diffusion and adsorption

To explore the impact of pore size on the ion diffusion and adsorption performance, we examined three distinct ZTC materials: AEI, BEC, and TUN. The AEI, BEC materials with medium microporous and three-dimensional pore channels are similar in structure with the TUN one. The obvious difference in the pore size are suitable to investigate the effect of pore size on ion adsorption and diffusion. Compared to TUN

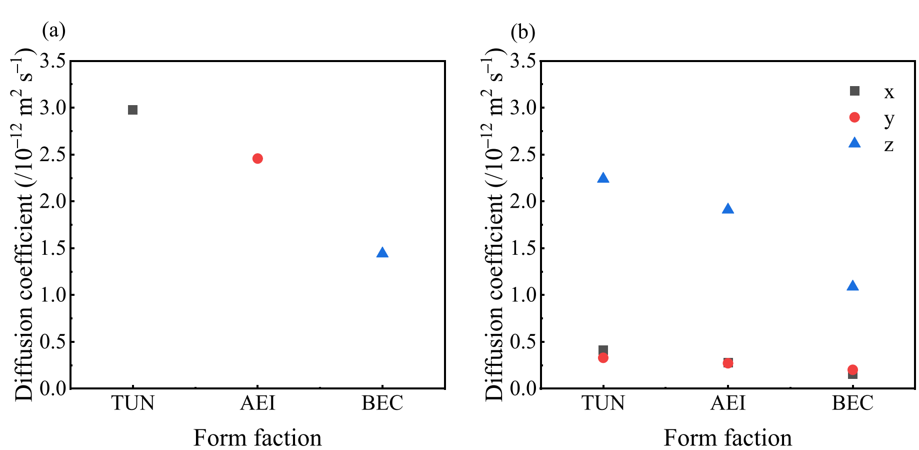

Fig. 13. Diffusion coefficient of ions in the three types of ZTC materials (a) the average value (b) the specific value in the x, y, and z-axis directions.

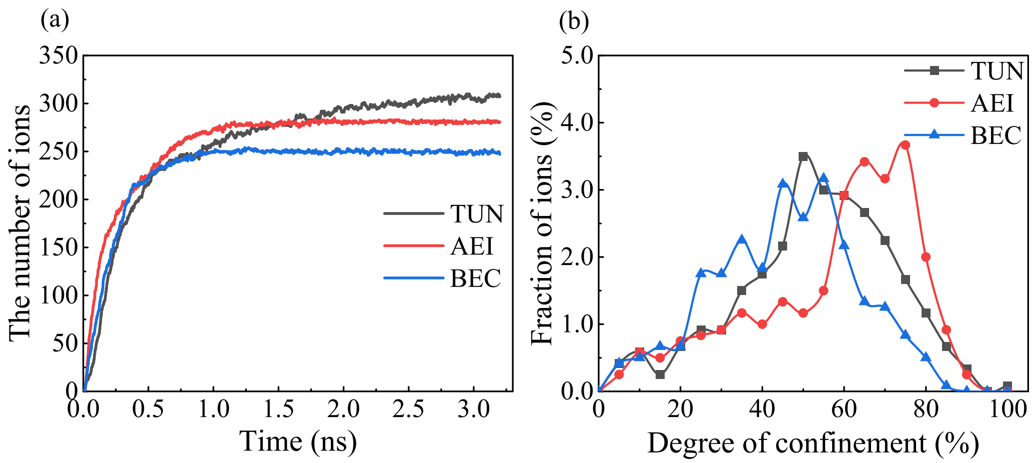

Fig. 14. (a) Time evolution of the number of ions adsorbed in the three materials, and (b) the DOC distribution of ions in the three materials.

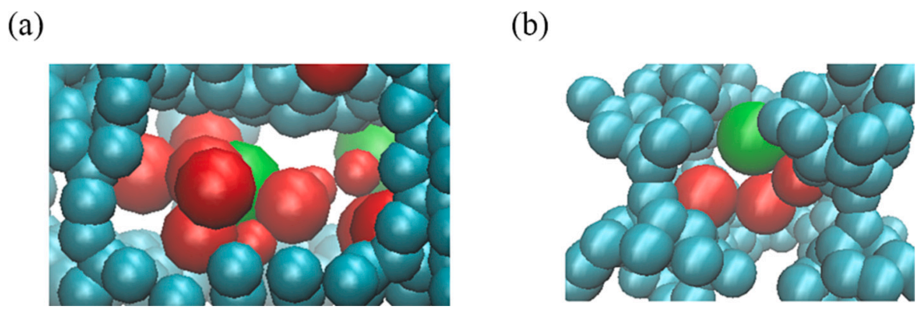

Fig. 15. Schematic diagram of (a) a half-enclosed ion, and (b) a fully confined ion.

materials, both AEI and BEC materials exhibit a more regular arrangement of carbon atoms and a denser structure. The basic structures and the pore size distribution diagram of these materials are depicted in Fig. 12. It is evident that the AEI material predominantly features a concentration of pores with a size of 5 Å, while the BEC material shows a concentration in the range of 7 to 8 Å. In contrast, the TUN material exhibits a bimodal distribution of pore sizes, accompanying with one more peak at 11–12 Å.

When Comparing the diffusion coefficient of ions across different pore structures as seen in Fig. 13, a correlation between ion diffusivity and pore size becomes apparent. In the case of carbon materials AEI and BEC, which possess concentrated pore size distributions, smaller pore sizes correspond to larger diffusion coefficients. Both the TUN and BEC materials exhibit peak pore size distributions in the range of 7 to 8 Å. However, the pore size distribution of the TUN material shows a bimodal pattern, including a pore size of 12 Å. Consequently, the ion diffusion coefficient in the TUN material surpasses that of the AEI and BEC materials. The complex structure of TUN-type ZTC materials, which includes variations in pore size and the number of connecting channels, becomes a predominant factor in determining diffusivity. This aligns with the earlier conjecture that the degree of adaptability with the carbon material directly impacts ion diffusivity. The pore structure of ZTC materials enhances the available space within the material and facilitates smoother diffusion channels. As a result, increased interionic distances during diffusion reduce electrostatic interactions, thereby promoting higher diffusivity.

Fig. 14(a) depicts the temporal variations of ion numbers in different ZTC materials. At the equilibrium state, the ion number variation curves in the AEI and BEC models show similarity, with the AEI material, accommodating a larger quantity of ions Notably, the TUN model possesses the highest number of ions among the three materials. This suggests that the complex pore structure of ZTC materials is more conducive to ion adsorption, enabling greater ion storage capacity. To provide a more precise depiction of the ion adsorption state, the distribution of DOC experienced by ions is calculated and shown in Fig. 14(b). It is clear that DOC mainly concentrates within a peak position range of 0.4 to 0.6. This is primarily due to the sparser carbon atom arrangement in the TUN and BEC structures, which prevents complete wrapping coverage of ions, resulting in predominantly semi-encapsulated ions. This observation is further supported by Fig. 15(a), which reveals that the majority of the ions diffusing into the TUN and BEC materials are concentrated in the central or edge regions of the void, with fewer ions found in the corners. Consequently, the concentration of DOC falls within the middle interval. In contrast, the denser carbon atom structure in AEI leads to a DOC peak position larger than 0.6, indicating a higher frequency of complete encapsulation of ions (as illustrated in Fig. 15(b)).

4. Conclusion

This study focuses on systematically conducting molecular dynamics simulations to investigate the diffusion and adsorption behavior of ionic liquids within porous carbon materials. The aim is to explore the interrelationships between aperture structure, ion size, pore size, and kinetic properties. Our findings demonstrate that the impact of anion size and pore size on the diffusion coefficient is not absolute. Although, in certain cases, diffusivity and the number of adsorbed ionic liquids in ZTC materials increase with larger ion and pore sizes, it is the alignment of ion size and pore size that primarily determines ion transport properties. Furthermore, the unique characteristics of ZTC materials promote the investigation of the effects of subtle differences in pore structure on ion diffusion and adsorption. The results reveal that materials with larger internal volumes exhibit faster ion transport, assuming a similar aperture structure. These findings on the transport characteristics of ionic liquids in ZTC materials offer valuable theoretical insights for future research involving the application of ZTC materials as electrodes.

CRediT authorship contribution statement

Fenhong Song: Conceptualization, Methodology, Visualization. **Ruifeng Chen:** Software, Data curation, Writing- Original draft preparation. **Jiaming Ma:** Investigation, Software, Data curation. **Xiwu Zhang:** Investigation, Reviewing. **Jing Fan:** Supervision, Writing- Reviewing and Editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests.

Data availability

Data will be made available on request.

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References

[1] J. Liu, J. Wang, C. Xu, Advanced energy storage devices: basic principles, analytical methods, and rational materials design, Adv. Sci. 5 (1) (2018) 1700322.
[2] P. Simon, Y. Gogotsi, Materials for electrochemical capacitors, Nat. Mater. 7 (11) (2008) 845–854.
[3] X. Feng, X. Shi, J. Ning, W. Dong, J. Zhang, Hao Yue, Z. Wu, Recent advances in micro-supercapacitors for AC line-filtering performance: from fundamental models to emerging applications, EScience 1 (2) (2021) 124–140.
[4] M.V. Fedorov, A.A. Kornyshev, Ionic liquids at electrified interfaces, Chem. Rev. 114 (5) (2014) 2978–3036.
[5] J.R. Miller, A. Burke, Electrochemical capacitors: challenges and opportunities for real-world applications, Electrochem. Soc. Interface 17 (1) (2008) 53.
[6] K. Jost, G. Dion, Y. Gogotsi, Textile energy storage in perspective, J. Mater. Chem. A 2 (28) (2014) 10776–10787.
[7] M.R. Lukatskaya, B. Dunn, Y. Gogotsi, Multidimensional materials and device architectures for future hybrid energy storage, Nat. Commun. 7 (1) (2016) 12647.
[8] W. Zhang, S. Wei, Y. Wu, Y. Wang, Poly (ionic liquid)-derived graphitic nanoporous carbon membrane enables superior supercapacitive energy storage, ACS Nano 13 (9) (2019) 10261–10271.
[9] O. Barbieri, M. Hahn, A. Herzog, R. Kötz, Capacitance limits of high surface area activated carbons for double layer capacitors, Carbon 43 (6) (2005) 1303–1310.
[10] M.F. DeVolder, S.H. Tawfick, R.H. Baughman, A. Johnhart, Carbon nanotubes: present and future commercial applications, Science 339 (6119) (2013) 535–539.
[11] J. Du, Y. Zhang, H. Lv, A. Chen, Re-assembly: construction of macropores in carbon sheets with high performance in supercapacitor, Adv. Powder Technol. 32 (4) (2021) 1294–1299.
[12] Z. Fan, J. Yan, L. Zhi, T. Wei, Q. Zhang, A three-dimensional carbon nanotube/graphene sandwich and its application as electrode in supercapacitors, Adv. Mater. 22 (33) (2010) 3723–3728.
[13] S. Li, G. Feng, P.F. Fulvio, C. Hillesheim, C. Liao, Molecular dynamics simulation study of the capacitive performance of a binary mixture of ionic liquids near an onion-like carbon electrode, J. Phys. Chem. Lett. 3 (17) (2012) 2465–2469.
[14] F. Bonaccorso, L. Colombo, G. Yu, M. Stoller, V. Tozzini, V. Pellegrini, Graphene, related two-dimensional crystals, and hybrid systems for energy conversion and storage, Science 347 (6217) (2015) 1246501.
[15] J. Hou, Y. Shao, M.W. Ellis, R.B. Moore, B. Yi, Graphene-based electrochemical energy conversion and storage: fuel cells, supercapacitors and lithium ion batteries, Phys. Chem. Chem. Phys. 13 (34) (2011) 15384–15402.
[16] Y. Yang, K.K. Ge, S. Rehman, B. Hong, Nanocarbon-based electrode materials applied for supercapacitors, Rare Metals 41 (12) (2022) 3957–3975.
[17] L. Liu, C. Hu, D. Nicholson, S.K. Bhatia, Inhibitory effect of adsorbed water on the transport of methane in carbon nanotubes, Langmuir 33 (25) (2017) 6280–6291.
[18] Q. Wang, L. Liu, C. Liu, J. Song, X. Gao, Size effect in determining the water diffusion rate in carbon nanotubes, J. Mol. Liq. 334 (2021), 116034.
[19] J. Song, L. Liu, Q. Li, C. Liu, F. Song, Entrance resistance of water transport into carbon nanotubes: insights from moleculardynamics simulations, J. Mol. Liq. 331 (2021) 115739.
[20] M. Moliner, C. Martínez, A. Corma, Multipore zeolites: synthesis and catalytic applications, Angew. Chem. Int. Ed. 54 (12) (2015) 3560–3579.
[21] H. Nishihara, T. Kyotani, Zeolite-templated carbons – three-dimensional microporous graphene frameworks, ChemComm 54 (45) (2018) 5648–5673.
[22] H. Nishihara, T. Kyotani, Zeolite-templated carbon – its unique characteristics and applications, Novel Carbon Adsorb. (2012) 295–322.
[23] E Lahrar .H., P. Simon, C. Merlet, Carbon–carbon supercapacitors: beyond the average pore size or how electrolyte confinement and inaccessible pores affect the capacitance, J. Chem. Phys. 155 (18) (2021).
[24] F. Borghi, C. Piazzoni, M. Ghidelli, P. Milani, A. Podestà, Nanoconfinement of ionic liquid into porous carbon electrodes, J. Phys. Chem. C 125 (2) (2021) 1292–1303.
[25] S. Zhang, M. Galiński, X. Liu, K. Sielicki, X. Chen, P.K. Chu, R. Holze, Investigation of the microstructure on the nanoporous carbon based capacitive performance, Microporous Mesoporous Mater. 310 (2021), 110629.
[26] E.H. Lahrar, I. Deroche, G.C. Matei, Simulations of ionic liquids confined in surface-functionalized nanoporous carbons: implications for energy storage, ACS Appl. Nano Mater. 4 (4) (2021) 4007–4015.
[27] L. Liu, D. Nicholson, S.K. Bhatia, Effects of flange adsorption affinity and membrane porosity on interfacial resistance incarbon nanotube membranes, ACS Appl. Mater. Interfaces 10 (40) (2018) 34706–34717.
[28] W. Jian, W. Zhang, X. Wei, B. Wu, W. Liang, Y. Wu, Engineering pore nanostructure of carbon cathodes for zinc ion hybrid supercapacitors, Adv. Funct. Mater. 32 (49) (2022) 2209914.
[29] E. Braun, Y. Lee, S.M. Moosavi, S. Barthel, R. Mercado, B. Smit, Generating carbon schwarzites via zeolite-templating, Proc. Natl. Acad. Sci. U. S. A. 115 (35) (2018) E8116–E8124.
[30] L. Sarkisov, A. Harrison, Computational structure characterisation tools in application to ordered and disordered porous materials, Mol. Simul. 37 (15) (2011) 1248–1257.
[31] M. Salanne, Ionic liquids for supercapacitor applications, Ionic Liquids II (2018) 29–53.
[32] C. Merlet, M. Salanne, B. Rotenberg, New coarse-grained models of imidazolium ionic liquids for bulk and interfacial molecular simulations, J. Phys. Chem. C 116 (14) (2012) 7687–7693.
[33] E.A. Crespo, N. Schaeffer, J.A.P. Coutinho, G.P. Sanchez, Improved coarse-grain model to unravel the phase behavior of 1-alkyl-3-methylimidazolium-based ionic liquids through molecular dynamics simulations, J. Colloid Interface Sci. 574 (2020) 324–336.
[34] L.I. Vazquez-Salazar, M. Selle, A.H. DeVries, S.J. Marrink, P.C.T. Souza, Martini coarse-grained models of imidazolium-based ionic liquids: from nanostructural organization to liquid–liquid extraction, Green Chem. 22 (21) (2020) 7376–7386.
[35] F. Song, B. Ma, J. Fan, B. Chen, B. Li, Molecular dynamics simulation on the electrowetting behaviors of the ionic liquid [BMIM][BF4] on a solid substrate, Langmuir 35 (30) (2019) 9753–9760.
[36] F. Song, J. Xue, B. Ma, J. Fan, Y. Wang, Y. Jiang, Wetting and electro-wetting behaviors of [Bmim][Bf4] ionic liquid droplet on lyophobic and lyophilic solid substrates, J. Mol. Liq. 347 (2022), 118405.
[37] F. Song, R. Chen, G. Wang, J. Fan, H. Niu, Coalescence and break-up behaviors of nanodroplets under AC electric field, Molecules 28 (7) (2023) 3064.
[38] D. Wolf, P. Keblinski, S.R. Phillpot, J. Eggebrecht, Exact method for the simulation of Coulombic systems by spherically truncated, pairwise r− 1 summation, J. Chem. Phys. 110 (17) (1999) 8254–8282.
[39] P.T. Aidan, H.M. Aktulga, B. Richard, D.S. Bolintineanu, W.M. Brown, P.S. Crozier, J. Pieter, A. Kohlmeyer, S.G. Moore, D.N. Trung, S. Ray, M.J. Stevens, T. Julien, T. Christian, S.J. Plimpton, LAMMPS - a flexible simulation tool for particle-based materials modeling at the atomic, meso, and continuum scales, Comput. Phys. Commun. 271 (2022), 108171.
[40] F. Song, H. Niu, J. Fan, Q. Chen, G. Wang, L. Liu, Molecular dynamics study on the coalescence and break-up behaviors of ionic droplets under DC electric field, J. Mol. Liq. 312 (2020), 113195.
[41] F. Song, F. Wang, J. Ma, J. Xue, J. Fan, Microstructure of ionic liquids mixed with water on the charged graphene surface: a coarse-grained molecular dynamics simulation study, J. Mol. Liq. 391 (2023), 123253.
[42] A. Striolo, The mechanism of water diffusion in narrow carbon nanotubes, Nano Lett. 6 (4) (2006) 633–639.
[43] J.H. Petropoulos, P.P. Roussis, Study of “non-Fickian” diffusion anomalies through time lags. IV. More complex cases of axial distance-dependent anomalies, J. Chem. Phys. 50 (9) (1969) 3951–3956.
[44] D.G. Levitt, Dynamics of a single-file pore: non-Fickian behavior, Phys. Rev. A 8 (6) (1973) 3050.
[45] J.A. VanMeel, L. Filion, C. Valeriani, D. Frenkel, A parameter-free, solid-angle based, nearest-neighbor algorithm, J. Chem. Phys. 136 (23) (2012), 234107.
[46] C. Merlet, C. Péan, B. Rotenberg, P.A. Madden, B. Daffos, Highly confined ions store charge more efficiently in supercapacitors, Nat. Commun. 4 (1) (2013) 2701.
[47] A. Eftekhari, Surface diffusion and adsorption in supercapacitors, ACS Sustain. Chem. Eng. (2018) 3692–3701.